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324733(24)

B. E. (Seventh Semester) Examination, April-May 2016

(New Scheme)

(Elect., EEE Engg. Branch)

ELECTRICAL DRIVES

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question.*

1. (a) Define Passive and Active Load Torque. 2

(b) A weight of 500 kg is being lifted up at a uniform

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speed of 1·5 m/s by a winch driven by a motor running at a speed of 1000 rpm. The moments of inertia of the motor and winch are 0·5 and 0·3 kg-m^2 respectively. Calculate the motor torque and the equivalent moment of inertia referred to the motor shaft. In the absence of weight, motor develops a torque of 100 N-m when running at 1000 rpm. 7

(c) Explain the steady-state stability of a drive depends on relative characteristics of the motor and load. 7

(d) A drive has following parameters : 7

J = 10 kg-m^2, T = 100 – 0·1 N, N-m. Passive load torque T_l = 0·05 N, N-m, where N is the speed in rpm. Initially the drive is operating in steady-state. Now it is to be reversed. For this motor characteristic is changed to T = –100 – 0·1 N, N-m. Calculate the time of reversal.

2. (a) What do you understand by constant-torque drive and constant power drive? 2

(b) Explain the operation of a closed-loop position control scheme. What are the roles of inner current control and speed control loops? 7

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(c) Half hour rating of a motor is 100 kW. Heating time constant is 80 min and the maximum efficiency occurs at 70% full load. Determine the continuous rating of the motor. 7

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(d) A motor operate on a periodic duty cycle consisting of a loaded period of 20 min and no load period of 10 min. The maximum temperature rise is 60°C heating and cooling time constant are 50 and 70 min respectively, when operating continuously on no load the temperature rise is 10°C determine : 7

(i) Minimum temperature during the duty cycle.

(ii) Temperature when the motor is loaded continuously.

3. (a) Why a dc series motor is more suited to deal with torque over loads than other dc motors? 2

(b) A 2-pole separately excited dc motor has the rating of 220 V, 100 A and 750 rpm. resistance of the armature is $0 \cdot 1\ \Omega$. The motor has two field coils which are normally connected in parallel. If is used to drive a load whose torque is expressed as

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$T_L = 500 - 0 \cdot 3\ N$, N-m where N is the motor speed in rpm. Calculate : 7

(i) The motor armature current and speed when the armature voltage is reduced to 110 V.

(ii) The motor speed and current when field coils are connected in series.

(c) Draw the speed torque characteristic for dynamic braking operation of dc series motor. Why torque becomes zero at finite speed. 7

(d) A dc series motor is feel from 600 V dc source through a chopper. The dc motor has the following parameters :

$r_a = 0 \cdot 04\ \Omega$, $r_s = 0 \cdot 06\ \Omega$, $K = 4 \times 10^{-3}$ Nm/amp^2.

The average armature current of 300 A is ripple free for a chopper duty cycle of 60% determine :

(i) Input power from the source.

(ii) Motor speed and motor torque. 7

4. (a) What is Single Phasing? 2

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(b) Derive an equivalent circuit for the dc dynamic braking of an induction motor and explain why it is necessary to account for the saturation in the Magnetic Circuit. 7

(c) Develop an expression in terms of motor parameters for the time required to plug an induction motor to 0·95 of synchronous speed in the reverse direction. 7

(d) Explain self controlled synchronous motor drives. 7

(a) Define the coefficient of Adhesion. 2

(b) A schedule speed of 45 Km/h is required between two stops 1·5 km apart. Find the maximum speed over the run if the stop is of 20 seconds duration. The values of acceleration and retardation are 2.4 km/h/s and 3·2 km/h/s respectively. Assume a simplified trapezoidal speed-time curve. 7

(c) Discuss 25 kV ac traction drive employing transformer with tap-changer. 7

(d) Describe the advantages and drawbacks of semiconductor converter controlled traction drives. 7

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